

Re-reading the *Blastocystis* SSU rRNA database

A curated re-reading of every *Blastocystis* SSU rRNA record on GenBank

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Code & data: <https://github.com/rytito/blastocystis-ssu-curation> (open source)

We all trust GenBank labels we have never audited

***Blastocystis* has changed identity:** once written off as a parasite, it now reads as a commensal, pathobiont at worst, associated with more diverse microbiomes and favorable metabolic profiles.

That raises the stakes for subtyping: health claims are subtype-resolved, and every subtype call inherits a GenBank label nobody audits

your sequence → nearest hit → that submitter's label → an earlier one's...

And the taxonomy is moving: ST10 became ST10 + ST42–44; new tentative subtypes appear in nearly every survey; field standards now ask for near-full-length data.

Piperni et al., Cell 2024 · Tito Tadeo & Stensvold, Trends Parasitol. 2025 · Tito Tadeo, Clin. Nutr. 2026 · Gentekaki et al., Trends Parasitol. 2026

What survives when one transparent rulebook is applied to the whole database?

Pipeline Architecture: Six Modules, One Rulebook

01

Acquisition

Single genus-wide Entrez query; 18,074 records fetched as full GenBank flat files

02

Filtering

Length filter 1,000–2,500 bp retains near-full-length sequences; 16,784 short records held for barcode analysis

03

Two-Stage Subtype Assignment

Stage I: explicit ST-tag parsing with confidence grading. Stage II: 8-mer Jaccard placement against curated seeds (tiers at 0.08 / 0.15 / 0.30)

04

Deduplication & Metadata

MD5 sequence digests collapse cross-label duplicates; hierarchical host coalescing; geo_loc_name extraction; technology inference

05

Novel Cluster Detection

Placement-rejected records clustered on 8-mer distance (cutoff 0.20, min size 2); run-specific novel_N labels assigned

06

Barcode Mapping & Disease Mining

Short records mapped by query-normalized k-mer containment; free-text fields classified against 10-category disease taxonomy

Scicluna, S. M., Tawari, B., & Clark, C. G., Protist. 2006 · Gentekaki et al., Trends Parasitol. 2026 · Maloney JG & Santin M., Microorganisms. 2021

All parameters recorded machine-readably in each run's `run_config.json`. Code available under MIT license:

github.com/rytito/blastocystis-ssu-curation

From 18,074 Records to a 1,052-Sequence Audited Reference Set

18,074

Records Retrieved

August 2026 GenBank snapshot

1,274

Near-Full-Length

1,000–2,500 bp; consistent with subtype-designation standards

222

Cross-Label Duplicates

Identical sequences deposited under conflicting subtype names

1,052

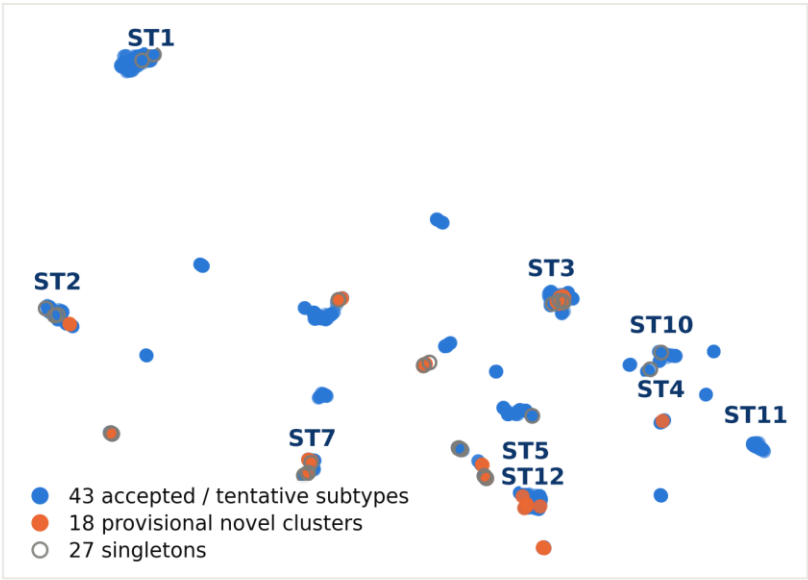
Final Reference Set

Spanning 62 labels; 92.7% confidently assigned

Subtypes are real, separable clusters, not just labels

Subtypes form discrete clouds

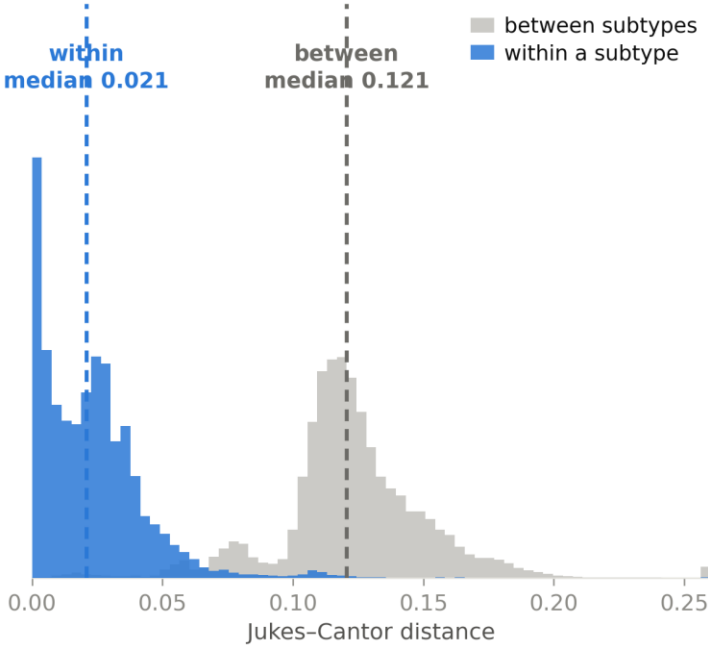
UMAP projection of the 1,052-record distance matrix



A 2-D projection — the quantitative test is at right.

...and the distances say the same

All 474,825 pairs among the 975 ST-labelled records



Between-subtype distances are 5.8× the within-subtype median. Cliff's $\delta = -0.981$ — near-total separation.

62 labels

in the final reference set

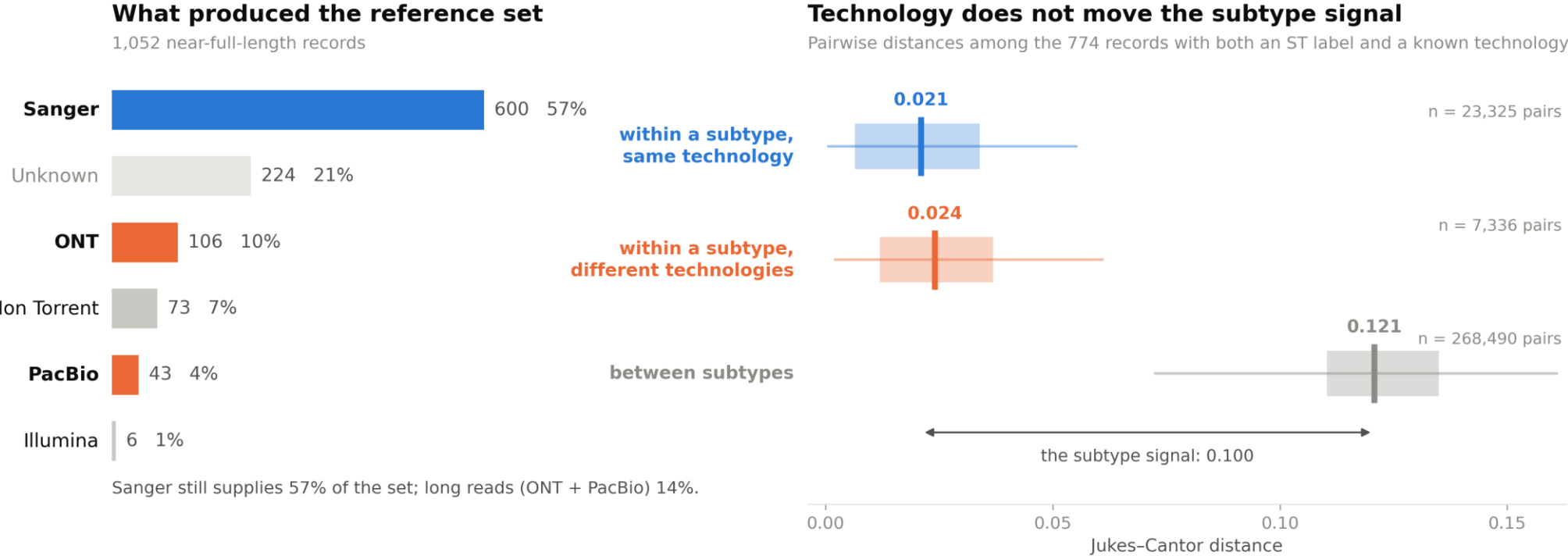
43 accepted or tentative subtypes (ST1-ST49)

18 provisional novel clusters

1 ST_unassigned — 27 singletons

975 of 1,052 records carry an accepted ST label.

Sanger-dominates and free of a technology confound

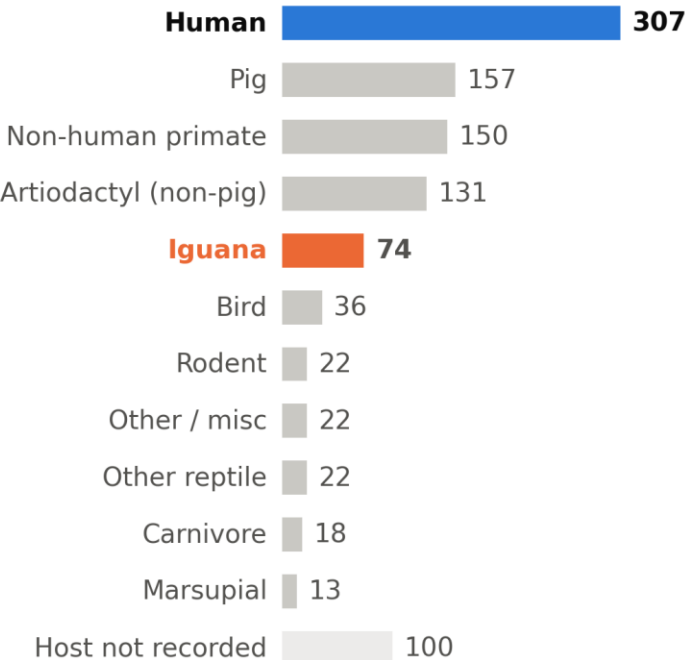


Mixing technologies shifts the within-subtype median by just 0.003 — 3% of the gap between subtypes (Cliff's δ = -0.13, negligible). No technology confound in subtype assignment.

Who has been sequenced: humans, pigs -> and one odd iguana batch

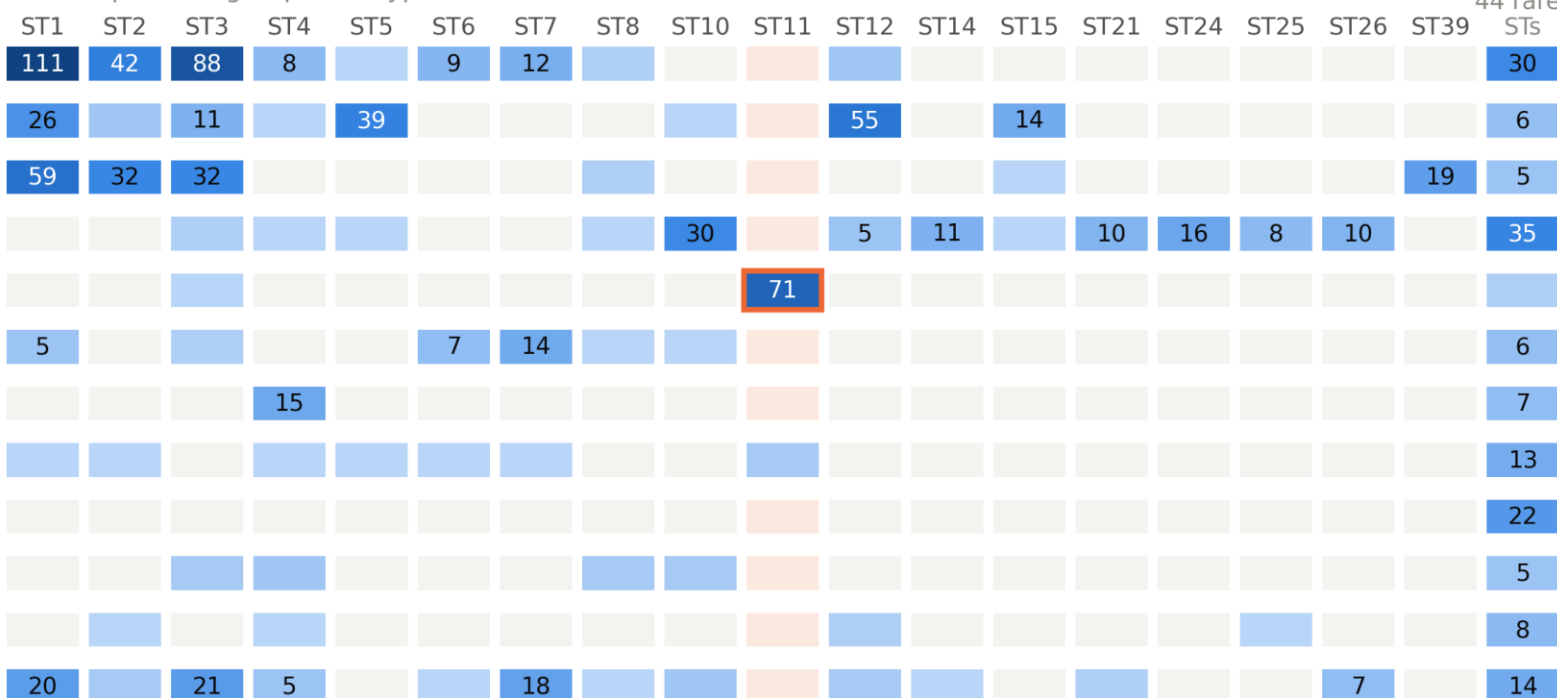
What got sequenced

952 host-annotated records



...and which subtypes they carry

Records per host group x subtype · shade scales within the matrix



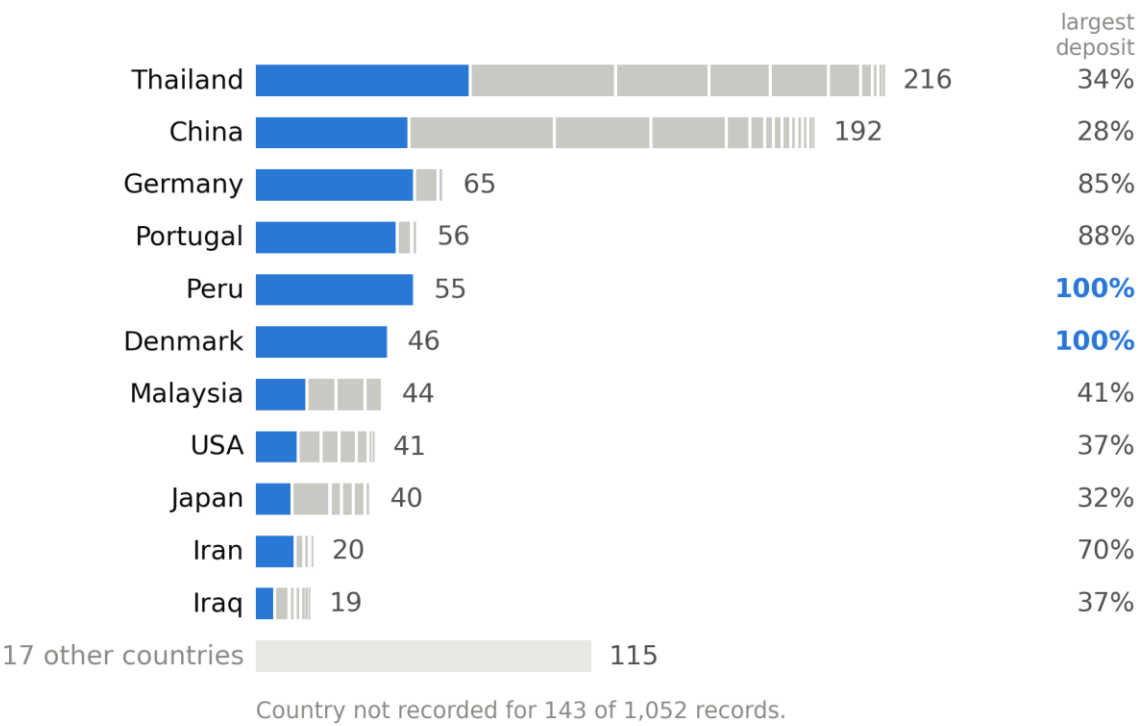
307 human records — 32% of the 952 with a host annotation. 71 of the 74 ST11 records are iguanas — and all 71 come from a single deposit (OQ693 · Thailand · 2022).

ST1 + ST3 dominate humans; ST5/ST12 the pigs; 71 of the 74 ST11 records are iguanas from one deposit — worth re-checking.

The map follows the sequencers, not the organism

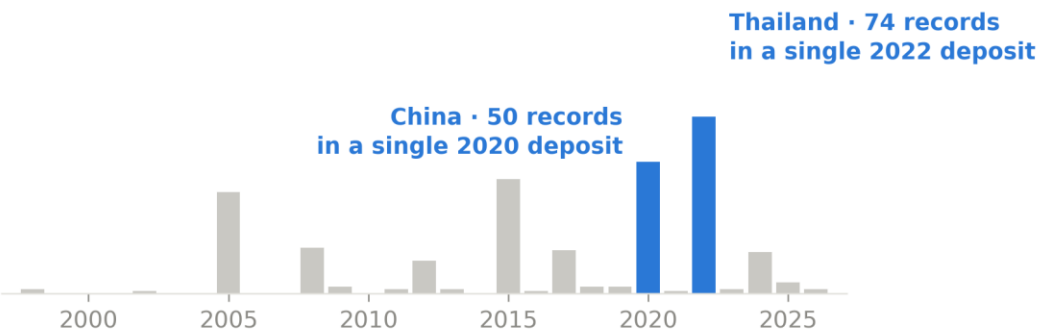
Where the records come from

Each bar split by submission batch · blue = the single largest deposit



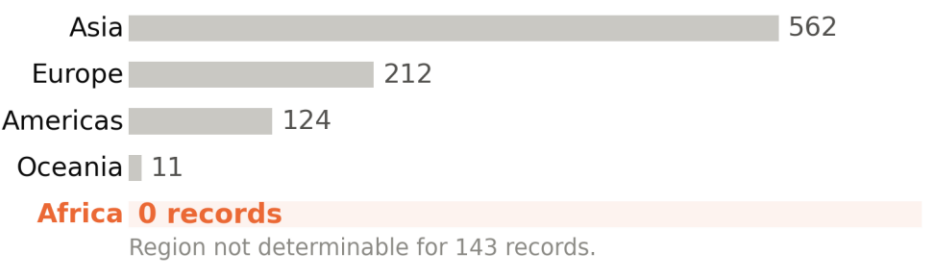
...and when they were collected

Only 345 of 1,052 records carry a collection date



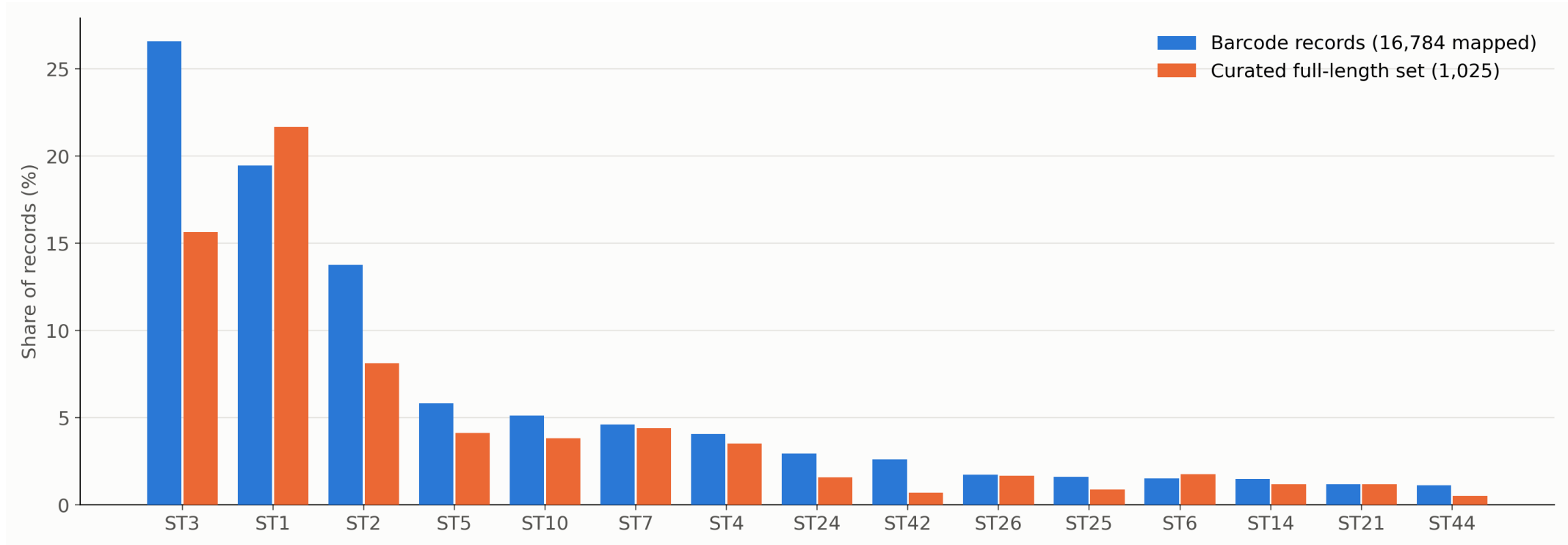
The gap that will still be here next year

29 countries are represented in the database. None of them is in Africa.



host	countries	collection	independent	records
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The barcode records, mapped back and they vote



92.8% of 16,784 barcode records get a confident subtype call against the curated references.

121 of them land in the novel clusters, independent support for the tortoise and beaver lineages.

Six tortoise species, two countries, one tight clade

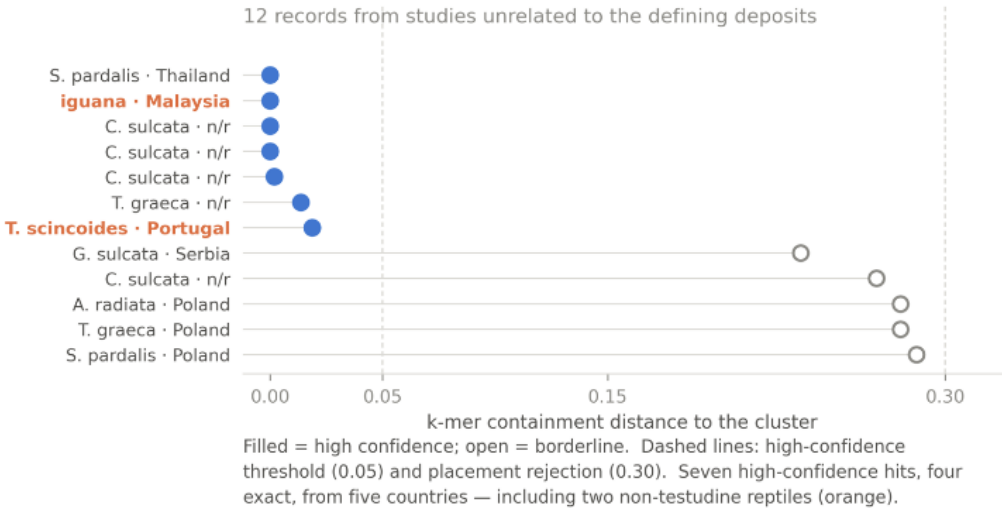


Full-length evidence (n = 7)

Six tortoise species, two countries, two unrelated submissions

host species	country	year	submission
● <i>Geochelone carbonaria</i>	Czechia	-	EF209
● <i>Agrionemys horsfieldii</i>	Japan	2012	KT438
● <i>Astrochelys radiata</i>	Japan	2012	KT438
● <i>Geochelone elegans</i>	Japan	2012	KT438
● <i>Geochelone elegans</i>	Japan	2012	KT438
● <i>Stigmochelys pardalis</i>	Japan	2012	KT438
● <i>Testudo graeca</i>	Japan	2012	KT438

Independent barcode records mapping to it



Median intra-cluster distance 0.006, tighter than the median accepted subtype (0.021), while rejecting placement against every accepted

Reptile Blastocystis is barely described, this looks like a genuine chelonian lineage, not a sampling artefact.

Stensvold, et al., Trends Parasitol. · 2007 · Gentekaki et al., Trends Parasitol. 2026

A hypothesis, not a subtype proposal -> the accessions are listed; -> anyone with reptile material can formalize it.

What this is — and what it is not

IT IS

A full audit of the database we all BLAST against

Traceable: every call carries its evidence

A generator of leads (tortoise, beaver, iguana)

IT IS NOT

A replacement for expert curation

A taxonomy proposal: novel clusters are hypotheses

A fix for the Africa gap: only new deposits fix that

What you can take home

CLINICAL & SURVEILLANCE

As *Blastocystis* turns biomarker of gut health, subtype calls carry new weight: and yours now trace back to an audited reference, checkable rather than inherited.

TAXONOMY & PHYLOGENY

A deduplicated, validated, full-length set to build the definitive tree from, re-run as deposits land.

WILDLIFE & ZOOZOSES

Ambiguous reptile or wildlife sequences? The novel clusters are ready-made hypotheses to test first.

EVERYONE

One shared, open reference across the clinical, veterinary and ecological corners of our field.

Curate once. Trace forever. Share with everyone.

18,074 GenBank records → 1,052 audited references → 18 candidate novel subtypes → 2 that look real.

The barcode world maps onto it. The tortoise lineage is yours to formalize (beavers too).

Pipeline, manifest and reference set:
github.com/rytito/blastocystis-ssu-curation

Thank you!

The screenshot shows the GitHub repository page for 'blastocystis-ssu-curation' by user 'rytito'. The repository is public and has 0 forks, 0 stars, and 0 watches. The main branch is 'main'. The repository contains a README.md, a LICENSE, a .gitignore, and several Python scripts and JSON files. The commit history shows 6 commits by rytito, with the most recent commit being 'Update README.md' 2 weeks ago. The repository is described as a 'Curation pipeline for all Blastocystis SSU rRNA records on GenBank — content-validated subtype assignment, novel-cluster detection, barcode mapping, full provenance'.

File	Description	Commit Date
results	Add disease-context mining (mine_disease_context.py) + ...	3 weeks ago
.gitignore	Blastocystis SSU rRNA curation pipeline v5	3 weeks ago
LICENSE	Blastocystis SSU rRNA curation pipeline v5	3 weeks ago
README.md	Update README.md	2 weeks ago
blastocystis_rRNA_pipeline_v5.py	Blastocystis SSU rRNA curation pipeline v5	3 weeks ago
config_talk_run_20260520.json	Blastocystis SSU rRNA curation pipeline v5	3 weeks ago
environment.yml	Blastocystis SSU rRNA curation pipeline v5	3 weeks ago
map_short_reads.py	Add barcode-region mapping analysis (map_short_reads....	3 weeks ago
mine_disease_context.py	Add disease-context mining (mine_disease_context.py) + ...	3 weeks ago

About
Curation pipeline for all Blastocystis SSU rRNA records on GenBank — content-validated subtype assignment, novel-cluster detection, barcode mapping, full provenance

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